INDUSTRIAL INTERNSHIP REPORT

STUDENT PERFORMANCE PREDICTION USING MACHINE LEARNING

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COLLEGE: APOLLO ENGINEERING COLLEGE

INTERNSHIP DOMAIN: DATA SCIENCE AND MACHINE LEARNING

Executive Summary

This internship focused on applying Data Science and Machine Learning concepts to predict student academic performance. The project demonstrates the complete machine learning workflow including problem definition, data collection, preprocessing, exploratory data analysis, feature engineering, model building, evaluation and prediction. Python along with Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn was considered for implementation. The proposed solution can assist educational institutions in identifying students who may require additional academic support. Throughout the internship I improved my practical knowledge of Python programming, data analysis, machine learning algorithms and technical documentation.

Preface

The internship provided an opportunity to connect academic concepts with practical applications. Working on this project helped me understand how data driven solutions can solve real world educational problems. The experience strengthened my analytical thinking, programming skills, documentation ability and confidence in working with datasets. I sincerely thank upskill Campus, UniConverge Technologies Pvt. Ltd. and Apollo Engineering College for providing this valuable learning opportunity.

Introduction

Data Science combines statistics, programming and domain knowledge to extract meaningful insights from data. Machine Learning enables computers to learn patterns from historical data and make predictions. In this project, student related attributes are analysed to estimate academic performance, enabling timely academic intervention.

PROBLEM STATEMENT

Educational institutions often find it difficult to identify students who are academically at risk before examinations. The objective of this project is to develop a machine learning model that predicts student performance using historical educational data such as attendance, study hours and assessment scores.

EXISTING AND PROPOSED SOLUTION

Traditional prediction methods rely mainly on manual observation and may not always be consistent. The proposed solution uses a machine learning model to provide faster, more consistent and data driven predictions that can support academic decision making.

METHODOLOGY

The project follows the stages of data collection, preprocessing, handling missing values, feature selection, train test split, model training using Random Forest, evaluation and prediction. Model performance is assessed using suitable evaluation metrics.

PERFORMANCE TESTING

The trained model is evaluated using previously unseen testing data. Performance is measured using accuracy and other standard evaluation measures to ensure reliable prediction results.

MY LEARNINGS

This internship improved my understanding of Python programming, data preprocessing, visualization, machine

learning concepts, model evaluation, GitHub usage and professional report preparation.

FUTURE SCOPE

Future improvements include larger datasets, advanced machine learning and deep learning models, deployment as a web application and integration with institutional academic management systems.